Research Report

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Summary of Revisions Since Draft 1

This page summarizes the main revisions made in response to feedback on the first draft.

- Revised the introduction to describe variables in substantive terms rather than primarily by code-style variable names.
- Clarified the data source, snapshot nature of the dataset, and observational scope of the study.
- Removed the report-body note about the local file path and kept that reproducibility detail in the R script instead.
- Expanded the data-preparation section to explain missing values, including why rows with missing prices were excluded and why the two observations missing fast-charging speed are unlikely to affect the fitted models materially.
- Added a note that extracted manufacturer names were manually checked after cleaning.
- Added justification that repeated model names appear to be distinct listings or configurations, so retaining them does not appear to directly violate the working independence assumption.
- Replaced the original price boxplot with a histogram to better display the distribution of vehicle prices.
- Revised the exploratory-analysis section so the price histogram, summary table, and price-versus-battery plot are the main figures tied to the research question.
- Added appendix figures showing the distributions of battery capacity and driving range by segment, price against additional technical variables, and broader pairwise relationships among quantitative variables.
- Added separate appendix diagnostic figures for the initial raw-price full model and the final log-linear model so that the justification for the log transformation and the adequacy of the final model are documented separately.
- Reduced the amount of detailed discussion of the simple exploratory regressions in Exploratory analysis part and shortened the correlation discussion so it functions mainly as a transition into the multiple-regression analysis.
- Revised the modeling-strategy section to use more formal wording, clarified that the initial raw-price model contains all covariates, and explained the backward variable-removal procedure more clearly.
- Reorganized the statistical-analysis section so the initial full model, the reason for the log transformation, the model-refinement step, and the final inferential model are presented in a clearer sequence with less repetition.
- Added a model-refinement table showing which terms were included in the initial full log-model and which term was removed before reaching the final model.
- Revised the final regression table to use a standard regression-output layout with estimates, standard errors, test statistics, and p-values.
- Strengthened the justification for the log transformation by linking it to the right-skewness and heteroscedasticity observed in the exploratory analysis and diagnostics.
- Revised the main-model-results section so the final table is interpreted in one continuous discussion rather than being split into a separate interpretation subsection.
- Expanded the final-model interpretation to explain the direction as well as the significance of the key coefficients, including the interaction term, the non-significant battery term, and the positive retained

control variables.

- Expanded the interpretation of the battery and range coefficients to explain why battery capacity is not statistically significant in the adjusted non-premium model and what that coefficient represents after controlling for related variables.
- Revised the interaction interpretation to focus more clearly on the difference between premium and non-premium pricing patterns.
- Added clearer references to appendix diagnostics, including separate references for the initial raw-price diagnostics and the final-model diagnostics, and explained the Cook's distance reference line.
- Revised the discussion and conclusion so they follow more naturally from the main model results, remove unnecessary references to simple models, and replace vague wording with more direct interpretation.

Introduction

Electric vehicle (EV) prices vary substantially across models, even among vehicles with similar technical specifications. Battery capacity is one of the most visible EV characteristics and is often associated with both production cost and consumer value. However, vehicle price is not determined by battery capacity alone. Manufacturers differ in brand positioning, and premium brands may charge systematically higher prices than non-premium brands even for vehicles with similar specifications. Understanding this relationship is important for both consumers and manufacturers. For consumers, it helps clarify whether higher prices reflect meaningful technical improvements or brand positioning. For manufacturers, it provides insight into how battery capacity is translated into pricing across different market segments.

This report studies EV models listed for sale in Germany and investigates whether battery capacity is associated with vehicle price, and whether that association differs between premium and non-premium manufacturers. The research questions are:

1. Among electric vehicle models listed for sale in Germany, is battery capacity positively associated with vehicle price?
2. Does the association between battery capacity and price differ between premium and non-premium manufacturers?

The data come from the *Electric Vehicle Specifications and Prices* dataset obtained from Kaggle and compiled by web scraping EV Database. EV Database aggregates manufacturer-reported technical specifications and listed prices for electric vehicle models listed for sale in Germany. The Kaggle dataset page indicates that the dataset entry was updated "2 years ago," which provides some background on when the posted dataset was last refreshed. However, that platform update note should not be interpreted as a precise statement of the model year coverage or the exact scrape date of the underlying EV listings. The dataset contains 360 observations, where each row represents a distinct EV model. This dataset should not be interpreted as a complete census of all electric vehicles that have ever existed or all EVs ever sold in Germany. Instead, it is best understood as a snapshot of the EV models included in EV Database when the source was scraped and the Kaggle dataset was assembled. The exact calendar year or scrape date is not stated in the raw CSV used here, so the report avoids treating the data as representing a single fully specified year. Since the data are observational rather than experimental, all conclusions in this report are associative rather than causal.

The response variable is the listed vehicle price in euros, which captures the manufacturer's suggested retail price for each electric vehicle in the German market. The main explanatory variable is battery capacity, measured in kilowatt-hours (kWh), which reflects the total energy storage available in the vehicle and is closely tied to both cost and performance. A second key variable is manufacturer segment, coded as premium or non-premium. Following the proposal, Tesla, BMW, Audi, Mercedes-Benz, and Porsche are classified as premium brands based on their widely recognized positioning in the European automotive market. This grouping is intended to reflect broad brand-tier differences rather than a formal regulatory classification. Other quantitative variables considered during model building include driving range, which measures how far a vehicle can travel on a single charge; energy efficiency, which captures how effectively the vehicle converts stored energy into distance traveled; top speed, the maximum speed the vehicle can achieve; acceleration, measured as the time in seconds required to go from 0 to 100 km/h; and fast-charging speed, which indicates how quickly the battery can be replenished using a rapid charging system. No major data transformations were applied to the variables. In particular, all quantitative variables were analyzed on their original scales. The only modification was to simplify the analysis by not distinguishing between specific vehicle models, and instead grouping vehicles at the manufacturer level through the premium versus non-premium classification.

This report uses multiple linear regression because the response is continuous and the main goal is inference about average relationships. In particular, including an interaction between battery capacity and market segment allows the battery-price relationship to differ between premium and non-premium manufacturers, which directly addresses the second research question. Variables named in the research questions are retained throughout model selection regardless of whether a selection procedure would otherwise remove them, as removing them would prevent direct interpretation of the parameters of primary interest.

Exploratory Analysis

Data Preparation

The raw file contains 360 observations and 9 recorded variables: `Battery`, `Car_name`, `Car_name_link`, `Efficiency`, `Fast_charge`, `Price.DE.`, `Range`, `Top_speed`, and `acceleration..0.100..` Because the data were scraped from an online database, the observed models reflect what EV Database included during the underlying collection period, not necessarily every EV available in the German market before or after that time. This matters for interpretation: the analysis is best understood as describing relationships among the EV models represented in this snapshot, rather than making claims about the broader German EV market.

Of the 9 variables, two contain missing values: the listed price in euros has 51 missing entries, and fast-charging speed has 2. Examination of the 51 observations with missing price reveals that they fall predominantly into identifiable categories — Chinese brands with limited official presence in the German market (such as Hongqi and XPENG), newly released models that had not yet received official German pricing at the time of data collection, commercial vehicles typically sold through fleet or quote-based arrangements, and ultra-luxury models for which public list prices are not routinely published. This pattern suggests the missing values are not random but are systematically related to market availability and pricing practices. Since price is the response variable, these 51 rows cannot contribute to the regression and are removed. The two observations with missing fast-charging speed are handled differently: they are retained in the working dataset and excluded only from models that include fast-charging speed as a predictor. Inspection of these two observations shows that their battery capacity and price fall close to the overall sample means, suggesting they are unlikely to be influential points, and their exclusion is not expected to meaningfully affect the fitted models.

To prepare the analysis dataset, a manufacturer variable was extracted from the vehicle name, the premium/non-premium grouping from the proposal was applied, the acceleration variable was renamed to a cleaner label, and only the variables relevant to the analysis were retained. Identifier variables such as `Car_name_link` were not used as predictors because they do not have an interpretable statistical relationship with price. Manufacturer names were extracted by taking the first word of the vehicle name. For multi-word brands such as Mercedes-Benz, a correction was applied explicitly in the code. The full list of extracted manufacturer names was manually reviewed, and no remaining misclassifications were identified in the dataset.

After removing rows with missing price, the main analysis dataset contains 309 EV models. A check of duplicate `Car_name` values shows that some names appear more than once. These rows were retained because they are not exact duplicates of the full record. In several cases, the repeated names correspond to different EV Database links and slightly different technical specifications, which suggests updated entries or distinct configurations rather than accidental duplication. For example, duplicated Mercedes entries often appear as an older version with missing price and a newer version with a reported price, while the duplicated Mini Cooper SE entries differ substantially in battery capacity and range and therefore represent genuinely different configurations. Because these repeated names appear to represent distinct listings rather than repeated measurements of the same vehicle record, retaining them does not appear to create a direct violation of the working independence assumption used in the regression analysis.

For the main regression, one practical choice is to fit models using complete cases on the variables included in each model. If `Fast_charge` remains in a candidate model, the effective sample size becomes 307 rather than 309 because of the two missing charging values. This should be noted when comparing models. To assess whether the 51 models with missing prices appeared systematically different from the retained models, the observed technical variables were compared across the two groups during data preparation. Those comparisons did not suggest large differences in battery capacity, range, efficiency, top speed, acceleration, or fast-charge capability, which reduces concern that the missing prices are concentrated only among very low-end or very high-end vehicles. However, because the missingness mechanism is not known, some selection bias remains possible.

Exploratory Summaries

Exploratory analysis serves two purposes in this study. First, it helps describe the data to readers who are not familiar with the EV market. Second, it helps identify features of the data that affect the regression analysis, such as skewness, segment differences, and multicollinearity among technical variables.

Start with summary statistics (Table 1) for the response and candidate explanatory variables.

Table 1: Summary statistics for key quantitative variables

N	Mean price	Median price	SD price	Mean battery	Median battery	SD battery	Mean range	Mean efficiency
309	67264.28	56942	34508.25	71.08	70.5	20.65	369.13	195.41

The 309 EV models used in the main analysis have a mean listed price of 67,264 EUR and a median price of 56,942 EUR, suggesting some right-skewness caused by a smaller number of high-priced vehicles (Table 1). Battery capacity is more balanced, with a mean of 71.1 kWh and a median of 70.5 kWh. Overall, the summary statistics indicate that price is more variable than battery capacity. Additional grouped summaries, not shown in Table 1, indicate that premium vehicles tend to have both larger batteries and higher listed prices on average than non-premium vehicles. These patterns motivate examining the joint relationship between price, battery capacity, and market segment in the regression analysis.

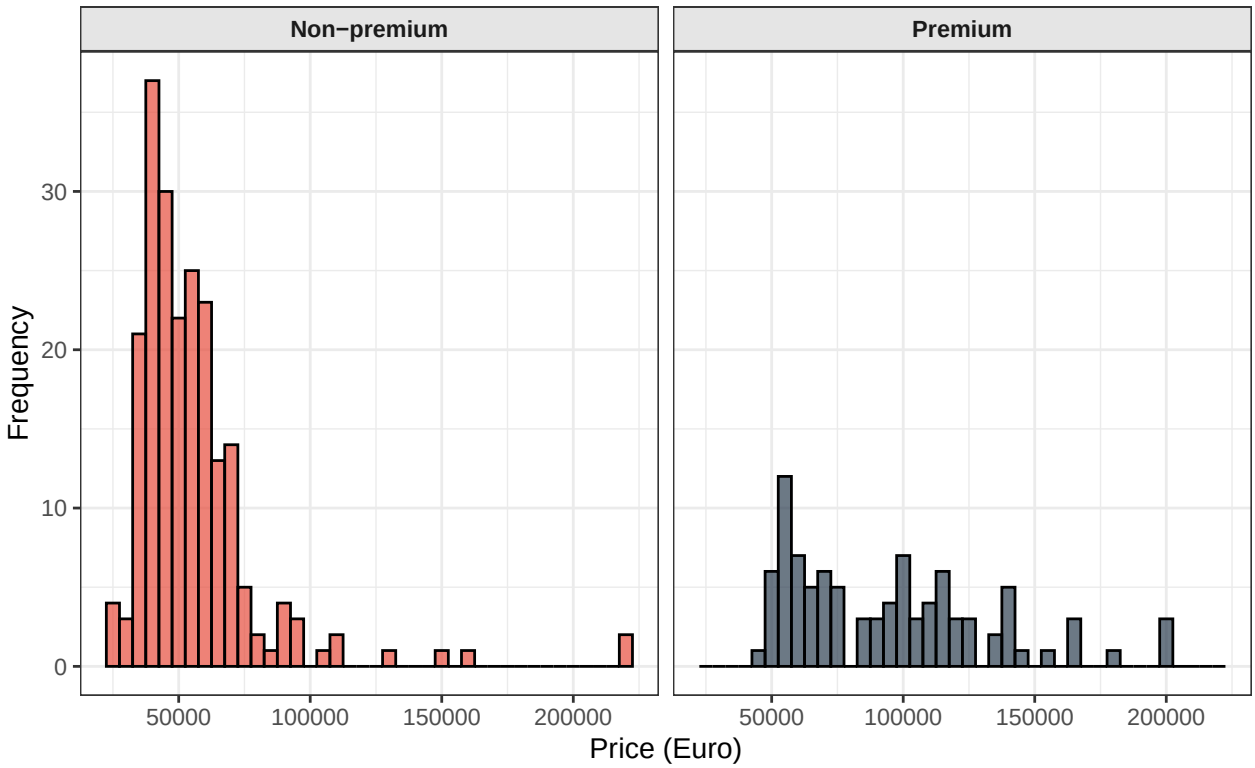


Figure 1: Distribution of EV prices by market segment

The histogram of vehicle prices shows clear right-skewness in both market segments (Figure 1). Most vehicles are concentrated in the lower-to-middle price ranges, with a smaller number of high-priced vehicles extending the upper tail. The premium segment is also more dispersed and shifted toward higher prices than the non-premium segment. These features are consistent with the summary statistics in Table 1 and suggest that a transformation of price may be useful in the regression analysis.

Other quantitative variables were also explored to understand how the two segments differ beyond price alone. Appendix Figure A1 shows the distributions of battery capacity and driving range by segment. Premium vehicles tend to have both larger batteries and longer ranges, and their distributions are more spread out than those of non-premium vehicles. Appendix Figure A2 complements this by plotting price against several additional technical variables, showing that price also tends to increase with range and top speed, while lower acceleration times are generally associated with higher prices. These broader exploratory patterns help motivate the later inclusion of multiple technical controls rather than focusing on battery capacity alone.

The main exploratory visualization plots price against battery capacity with separate fitted lines for premium and non-premium manufacturers (Figure 2). This figure directly addresses both research questions by showing both the overall battery-price association and the extent to which that association may differ by market segment.

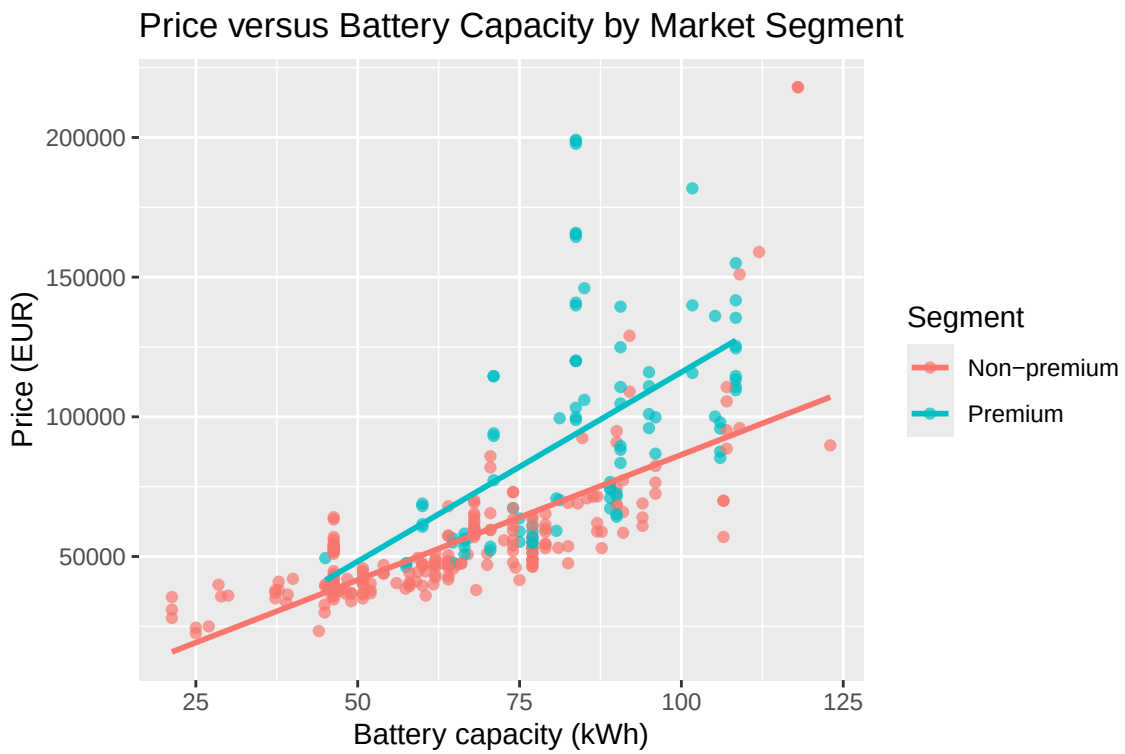


Figure 2: Price versus battery capacity by market segment

To complement Figure 2, Table 2 summarizes simple within-segment slopes from separate regressions of price on battery capacity. The estimated relationship is positive in both segments and steeper for premium vehicles, which is consistent with the visual pattern in Figure 2. Because these simple regressions are included only for exploratory comparison, the main inferential conclusions are based on the multiple-regression models developed in the next section.

Table 2: Estimated battery-price relationship by market segment

Segment	Slope estimate (EUR/kWh)	Standard error	p-value
Non-premium	896.68	57.47	<0.001
Premium	1355.14	222.63	<0.001

Because several technical variables may measure overlapping aspects of vehicle performance, their pairwise

correlations were also examined (Table 3). In particular, battery capacity is highly correlated with driving range (0.88), and top speed is strongly negatively correlated with acceleration time (-0.84). These patterns suggest that some predictors contain overlapping information, which is important to keep in mind when fitting and interpreting the multiple-regression model. Broader pairwise relationships are shown in Appendix Figure A3.

Table 3: Correlation matrix for quantitative variables

	Price.DE.	Battery	Range	Efficiency	Top_speed	Acceleration_0_100	Fast_charge
Price.DE.	1.00	0.70	0.59	0.17	0.76	-0.53	0.62
Battery	0.70	1.00	0.88	0.16	0.70	-0.61	0.59
Range	0.59	0.88	1.00	-0.31	0.76	-0.75	0.72
Efficiency	0.17	0.16	-0.31	1.00	-0.21	0.42	-0.29
Top_speed	0.76	0.70	0.76	-0.21	1.00	-0.84	0.79
Acceleration_0_100	-0.53	-0.61	-0.75	0.42	-0.84	1.00	-0.68
Fast_charge	0.62	0.59	0.72	-0.29	0.79	-0.68	1.00

Statistical Analysis

Modeling Strategy

The study employs a multiple linear regression framework because the response variable is continuous and the goal is inference about average relationships. The model is built around the research-question terms: battery capacity (*Battery*), market segment (*market_segment*), and their interaction. These terms are retained throughout model selection so that both research questions remain directly addressable. Additional technical variables, including *Range*, *Efficiency*, *Top_speed*, *Acceleration*_0_100, and *Fast_charge*, are treated as additional candidate variables considered during model selection.

Backward elimination is used for the additional candidate variables only, implemented with `drop1()` in R. Starting from the full model, variables with large p-values (> 0.05) are removed one at a time while the research-question terms are kept in the model. In particular, battery capacity, market segment, and their interaction remain in the model throughout selection because they are needed to address the two research questions directly. This approach is appropriate here because the project focuses on inference rather than prediction. It also keeps the model easier to interpret while preserving the terms needed to answer the main research questions.

Initial Full Model

Table 4: Summary of the initial raw-price model

Measure	Summary
Response scale	Price in EUR
Model content	Full model with all candidate covariates
Observations used	307
Observations dropped	2
Adjusted R-squared	0.7313
Residual standard error	17865
Overall model p-value	4.2e-82

Table 4 summarizes the initial raw-price model, meaning the model fitted on the original price scale with all candidate covariates included. Specifically, it contains the three research-question terms (battery capacity, market segment, and their interaction) together with five additional technical variables: driving range, energy

efficiency, top speed, acceleration, and fast-charging speed. Because `Fast_charge` is part of this full model, the two observations with missing fast-charging speed are omitted automatically, leaving 307 observations for estimation. This model explains a substantial share of the observed variation in price (adjusted $R^2 = 0.7313$), but that alone is not enough to justify interpretation. It serves mainly as the starting point for diagnostics and model refinement rather than as the final inferential model.

Model Refinement and Selection

Figure 1 shows that price is right-skewed, especially in the premium segment. Consistent with that pattern, Appendix Figure A4 shows that the raw-price full model has increasing residual spread in the residuals-versus-fitted plot and a heavy upper tail in the Q-Q plot. Together, these features indicate that the model on the original price scale does not satisfy the constant-variance and approximate-normality assumptions well. For that reason, a natural log transformation of price was used before model refinement. Starting from the same full set of terms, the additional technical variables were then checked one at a time and removed only when their evidence was weak, while battery capacity, market segment, and their interaction were kept in the model throughout because they are needed to answer the two research questions directly.

Table 5 summarizes how each term changed from the initial log-linear full model to the final model.

Table 5: Terms retained or removed during model refinement

Term	Initial full log-model p-value	Final model status
Battery capacity	0.1000	Retained
Premium segment	0.0065	Retained
Battery capacity x premium segment	<0.001	Retained
Driving range	0.0028	Retained
Energy efficiency	<0.001	Retained
Top speed	<0.001	Retained
Acceleration (0-100 km/h)	0.3263	Removed
Fast-charging speed	<0.001	Retained

It shows that all three primary terms tied to the research questions were retained throughout model refinement (Table 5). Among the five additional technical variables, only acceleration was removed, while driving range, energy efficiency, top speed, and fast-charging speed remained in the final model. The initial full-model p-values are included only to show which additional variables looked weakest before refinement; the final coefficient estimates and final-model p-values are reported separately in Table 6. The log transformation also improved overall fit, with the adjusted R^2 increasing from 0.7313 in the initial full model to 0.8412 in the final log-linear model. Although some mild departures remain, the diagnostic plots in Appendix Figure A5 and the Cook's distance values in Appendix Figure A6 do not indicate problems severe enough to rule out the final log-linear model for inference.

Main Model Results (log linear)

The final log-linear model was fitted on 307 observations; the two rows with missing fast-charging speed are omitted because fast-charging speed is retained in the model. Table 6 reports the final coefficient estimates, standard errors, test statistics, and p-values for the retained terms.

As shown in Table 6, the main result is the positive and highly significant interaction between battery capacity and premium segment ($\beta = 0.0051$, $p < 0.001$). This is the key evidence for the second research question: after the other retained technical variables are held fixed, the battery-price relationship differs between premium and non-premium vehicles. In plain terms, larger batteries are more closely reflected in the prices of premium vehicles than in the prices of non-premium vehicles. The coefficient for `market_segmentPremium` is also statistically significant ($p = 0.0078$), but because the model includes an interaction between battery capacity and market segment, this coefficient represents the segment difference when battery capacity is zero.

Since a battery capacity of zero kWh is not meaningful in this context, the coefficient is not interpreted on its own. Instead, the interaction term and the fitted relationship across battery levels are more substantively relevant.

Table 6: Coefficient estimates and p-values for the final regression model

	Estimate	Std. Error	t value	p-value
(Intercept)	8.0131	0.2184	36.6899	<0.001
Battery	-0.0047	0.0033	-1.4371	0.1517
market_segmentPremium	-0.3002	0.1120	-2.6802	0.0078
Range	0.0018	0.0006	2.8726	0.0044
Efficiency	0.0075	0.0010	7.4841	<0.001
Top_speed	0.0057	0.0005	10.5893	<0.001
Fast_charge	0.0003	0.0001	3.7778	<0.001
Battery:market_segmentPremium	0.0051	0.0013	3.7781	<0.001

Table 6 also shows that the coefficient for battery capacity in the baseline non-premium segment is negative, but not statistically significant ($p = 0.1517$). This does not make battery capacity irrelevant; rather, it means that once driving range and other related performance measures are already in the model, battery capacity by itself does not show a clear additional association with price for non-premium vehicles. This is plausible because battery capacity is strongly correlated with driving range ($r = 0.88$, Table 3), so much of the simple battery-price pattern is shared with other technical measures. In practical terms, once range is already in the model, the battery coefficient asks whether two vehicles with the same range but different battery capacities still differ in price, which is a much narrower question than the simple battery-price relationship. The retained control variables also matter: the coefficients for range, efficiency, top speed, and fast-charging speed are all positive, and each is statistically significant, indicating that vehicles with longer range, higher efficiency values, higher top speed, and faster charging tend to have higher prices after the other retained terms are held fixed. This pattern shows that EV prices reflect a broader bundle of performance characteristics rather than battery capacity alone. This is also why acceleration was removed during refinement: among the additional technical variables, it showed the weakest evidence once the other performance-related terms were already in the model.

Discussion and Conclusion

Discussion

Taken together, the results suggest that EV prices are shaped by a bundle of related technical characteristics rather than by battery capacity alone. In the non-premium segment (the baseline group), battery capacity does not show a clear independent association with price once other performance variables are included, suggesting that its effect is largely captured through measures such as range.

In contrast, the positive and significant interaction term indicates that the association between battery capacity and price is stronger in the premium segment. Because the main battery coefficient represents the effect in the non-premium segment, the interaction term captures how this relationship changes for premium vehicles. Together, these results indicate that the battery–price relationship is weak in the non-premium segment but becomes stronger in the premium segment. More specifically, battery capacity is associated with a larger increase in price for premium vehicles, suggesting that increases in battery size translate more directly into higher prices in this segment, even after controlling for other technical characteristics.

Overall, these findings indicate that EV pricing reflects a combination of performance attributes, and that the importance of battery capacity varies across market segments. One possible explanation is that premium consumers may place additional value on battery capacity itself, beyond the performance outcomes it enables.

Limitations

Several limitations should be kept in mind. First, the dataset is an observational snapshot of EV models listed in EV Database and does not represent all EVs sold in Germany or all EVs from a clearly specified year. Second, the premium versus non-premium classification is a simplified grouping based on widely recognized market positioning rather than a formal industry standard. Third, some relevant determinants of price, such as interior quality, software features, or driver-assistance packages, are not recorded in the dataset. Finally, because 51 observations had missing prices and were excluded, some risk of selection bias remains even though the missing-price models do not appear dramatically different on the observed technical variables.

Conclusion

The analysis provides evidence that battery capacity is associated with EV price, but that the relationship differs by market segment. After adjustment for other retained technical variables, battery capacity does not have a clear independent association with price for non-premium vehicles, whereas the significant positive interaction term shows that the battery-price relationship remains stronger in the premium segment. Overall, the findings suggest that EV pricing reflects both battery capacity and broader performance characteristics, and that these relationships are not the same across manufacturer segments.

Statement of AI Use

Generative AI was used to help organize the structure and wording of this report and to assist in preparing a separate reproducible analysis script. The statistical conclusions, numerical results, and model interpretations are based on outputs generated from the dataset itself.

Representative prompts used:

1. "Help revise the writing so the report reads like a formal statistical report rather than notebook-style notes."
2. "Help update the data-preparation description so it accurately reflects the raw dataset, missing values, and derived variables used in the analysis."
3. "Help strengthen the wording in response to feedback on interaction interpretation, diagnostics, model selection, and limitations."
4. "Help move less essential material to the appendix so the main body remains concise and within the page limit."
5. "Help revise the discussion and conclusion so they align with the final fitted model and avoid vague wording."
6. "Help update table and figure references so the report reads more like a polished paper and less like an analysis notebook."

Appendix

Supplementary Figures

Figure A1

Distributions of Battery Capacity and Driving Range by Market Segment

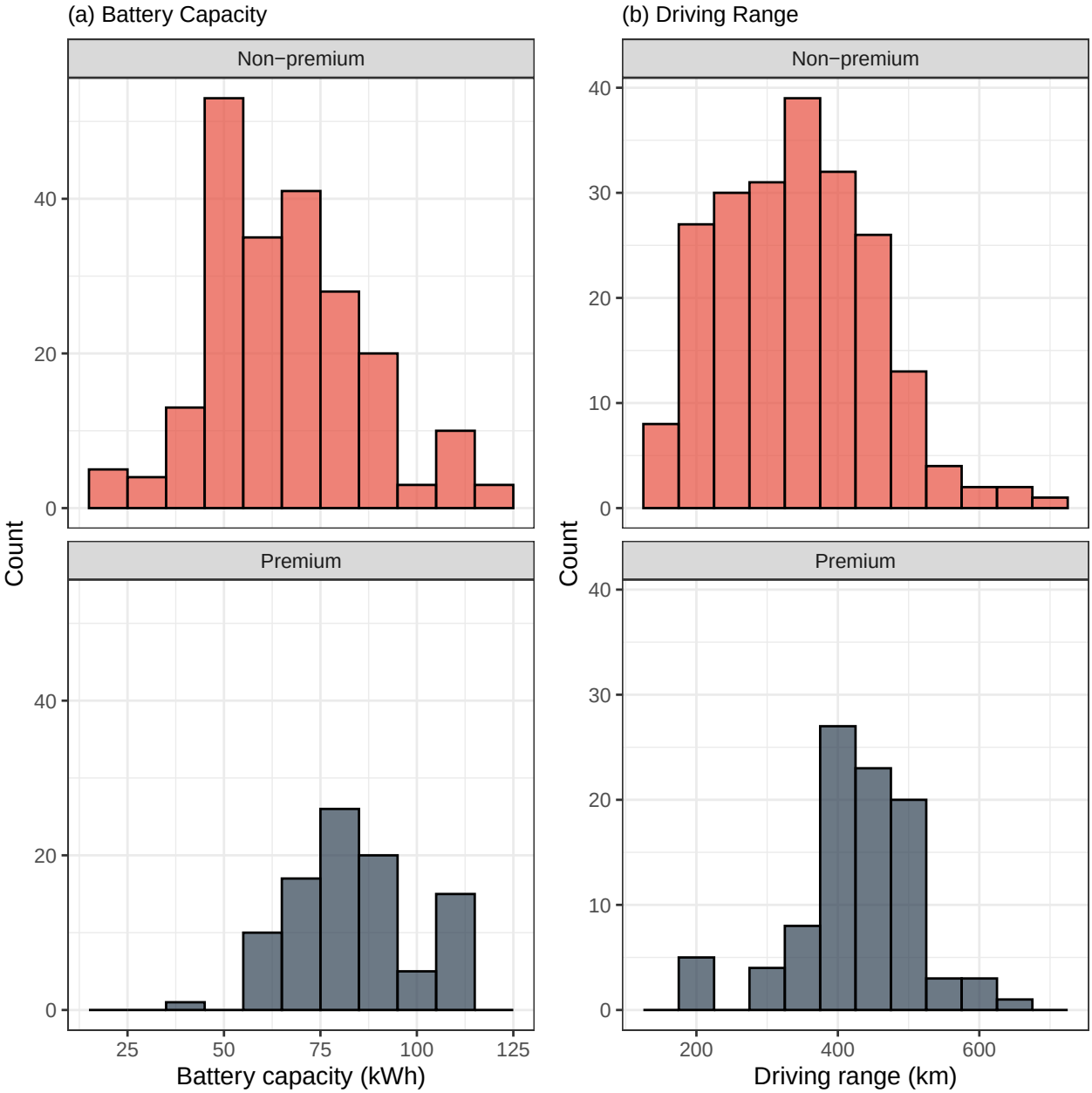


Figure A2

Price versus Additional Technical Variables by Market Segment

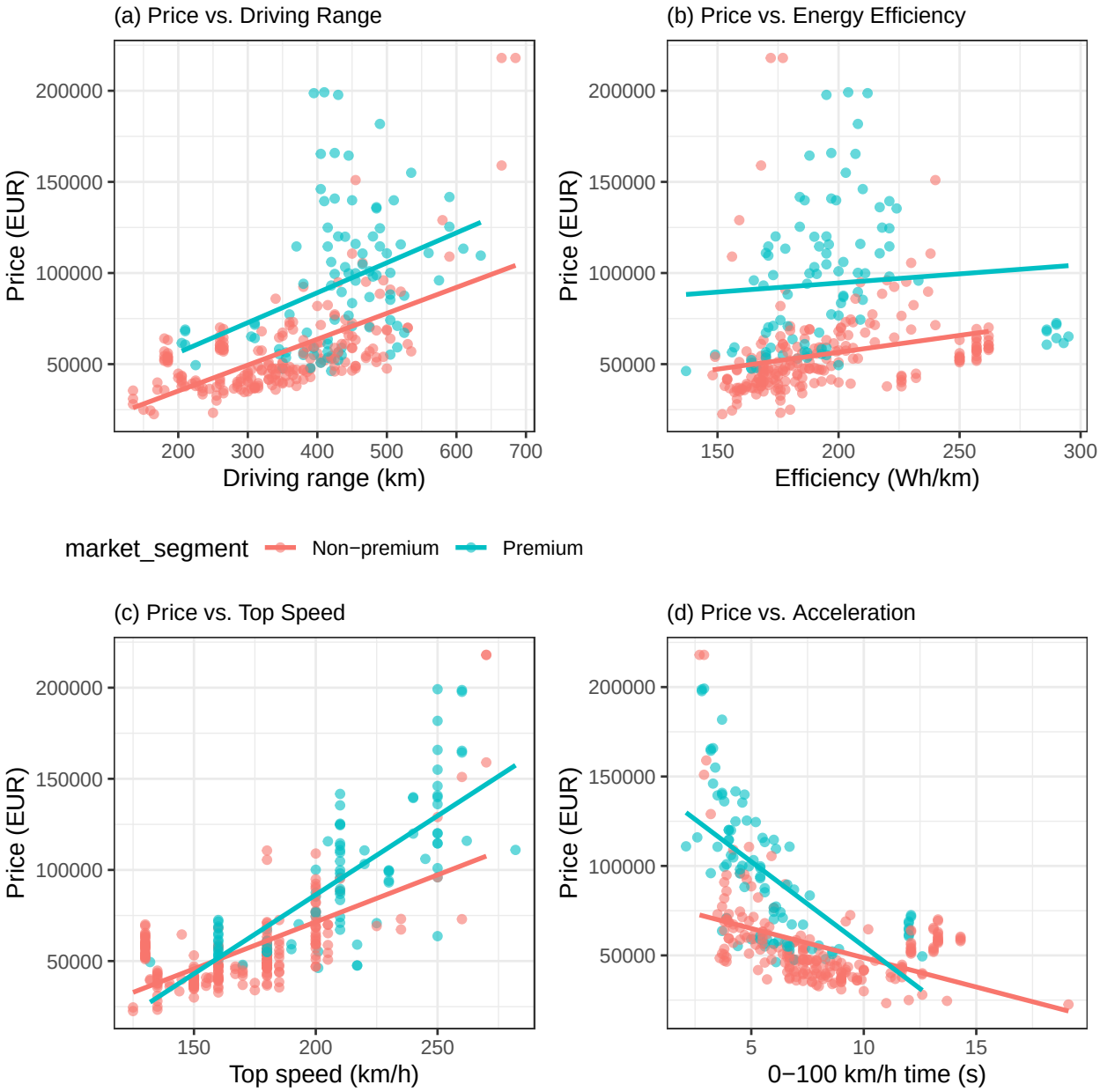
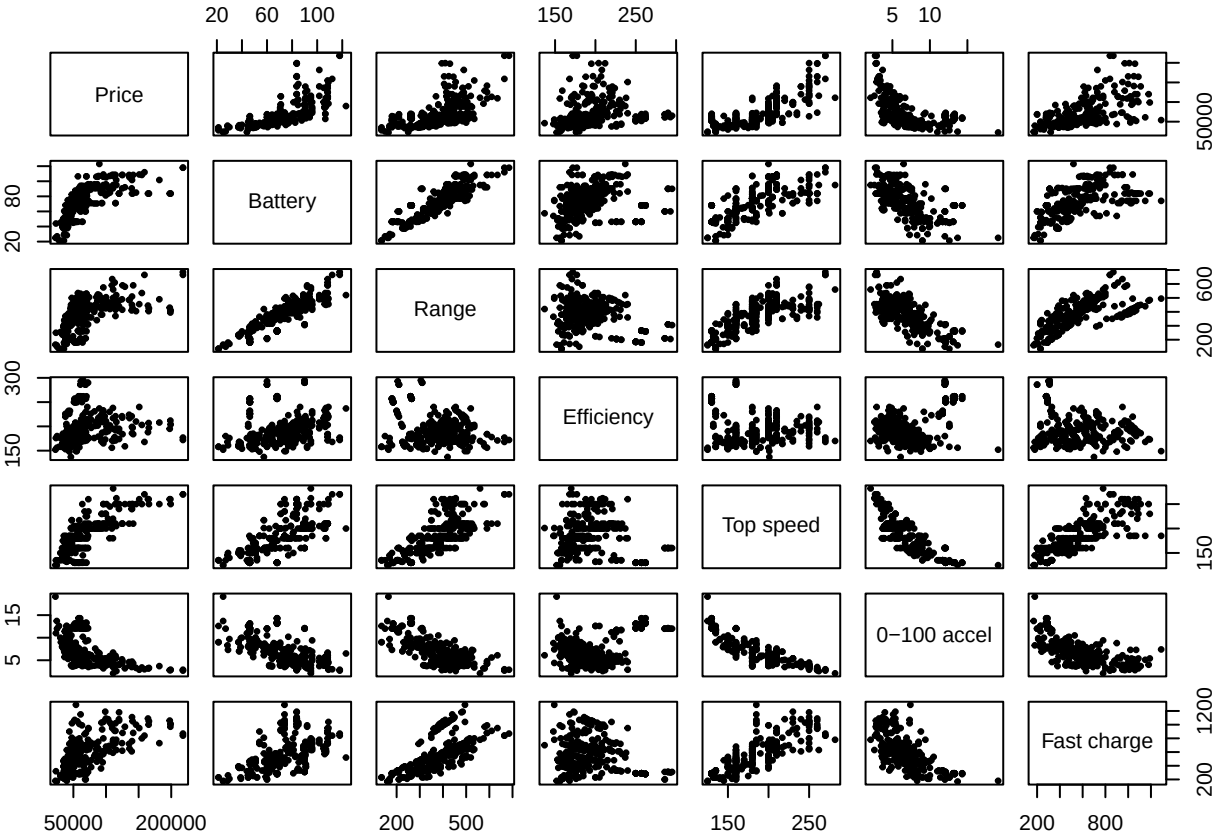


Figure A3

Scatterplot Matrix for Quantitative Variables Used in the Analysis



The scatterplot matrix in Appendix Figure A3 is mainly helpful for visually checking which explanatory variables appear strongly associated with price and which predictors appear highly related to one another. In this dataset, it supports the conclusion that variables such as `Battery`, `Range`, `Top_speed`, and `Fast_charge` share substantial overlap, which helps explain why a simpler final model is preferred.

Figure A4

Diagnostic Plots for the Initial Raw-Price Full Model

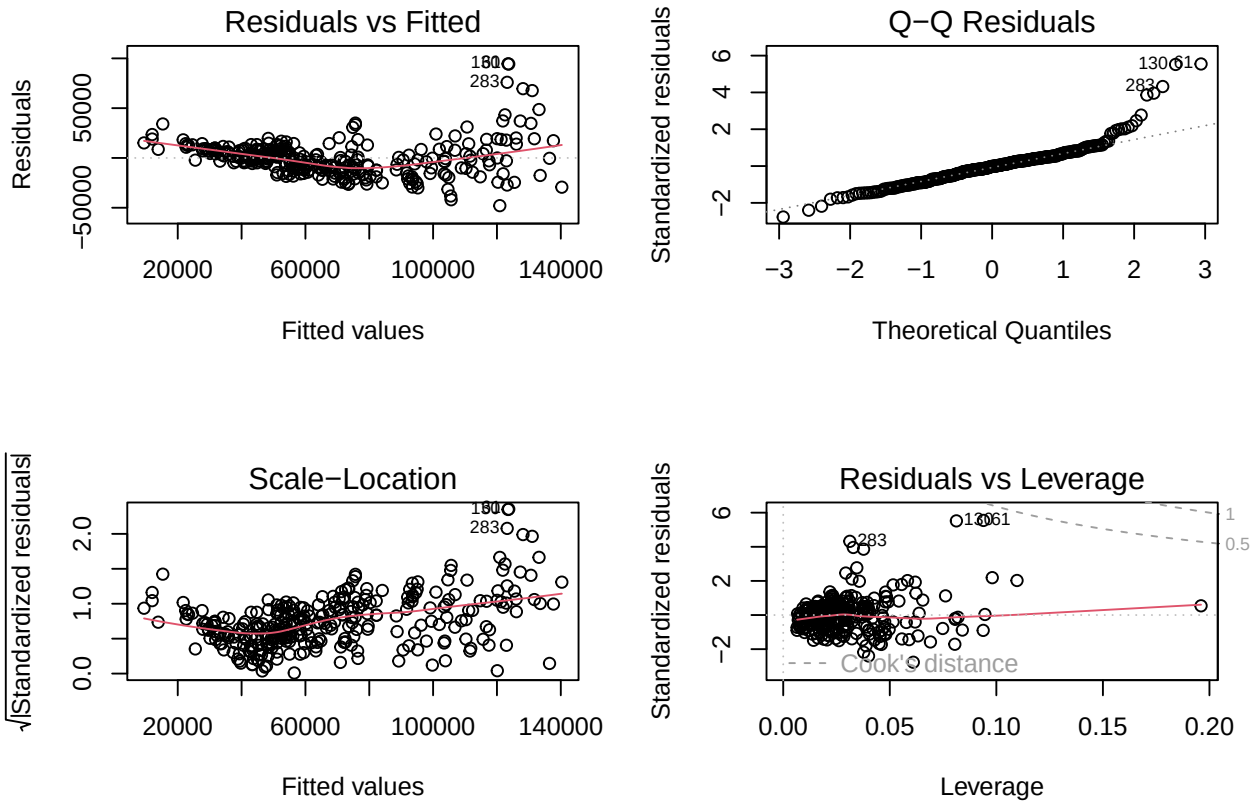


Figure A5

Diagnostic Plots for the Selected Regression Model

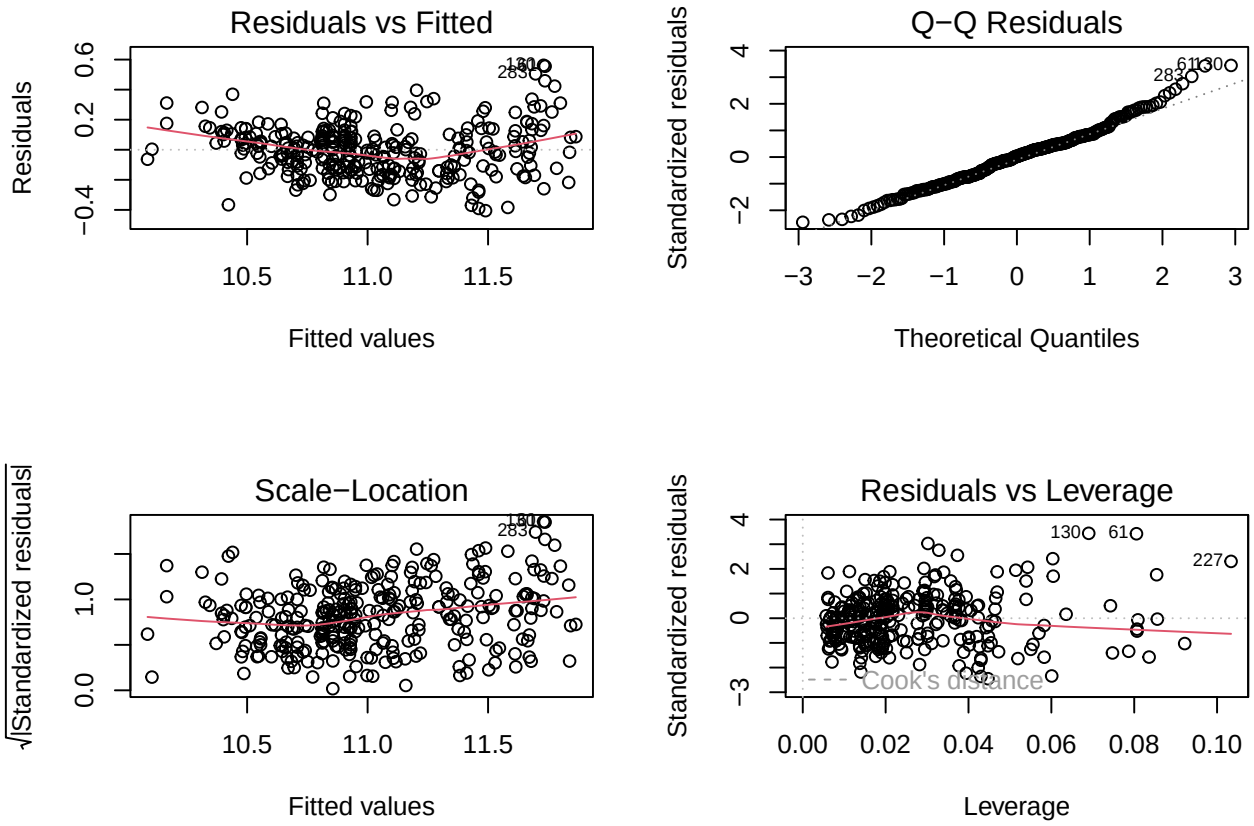
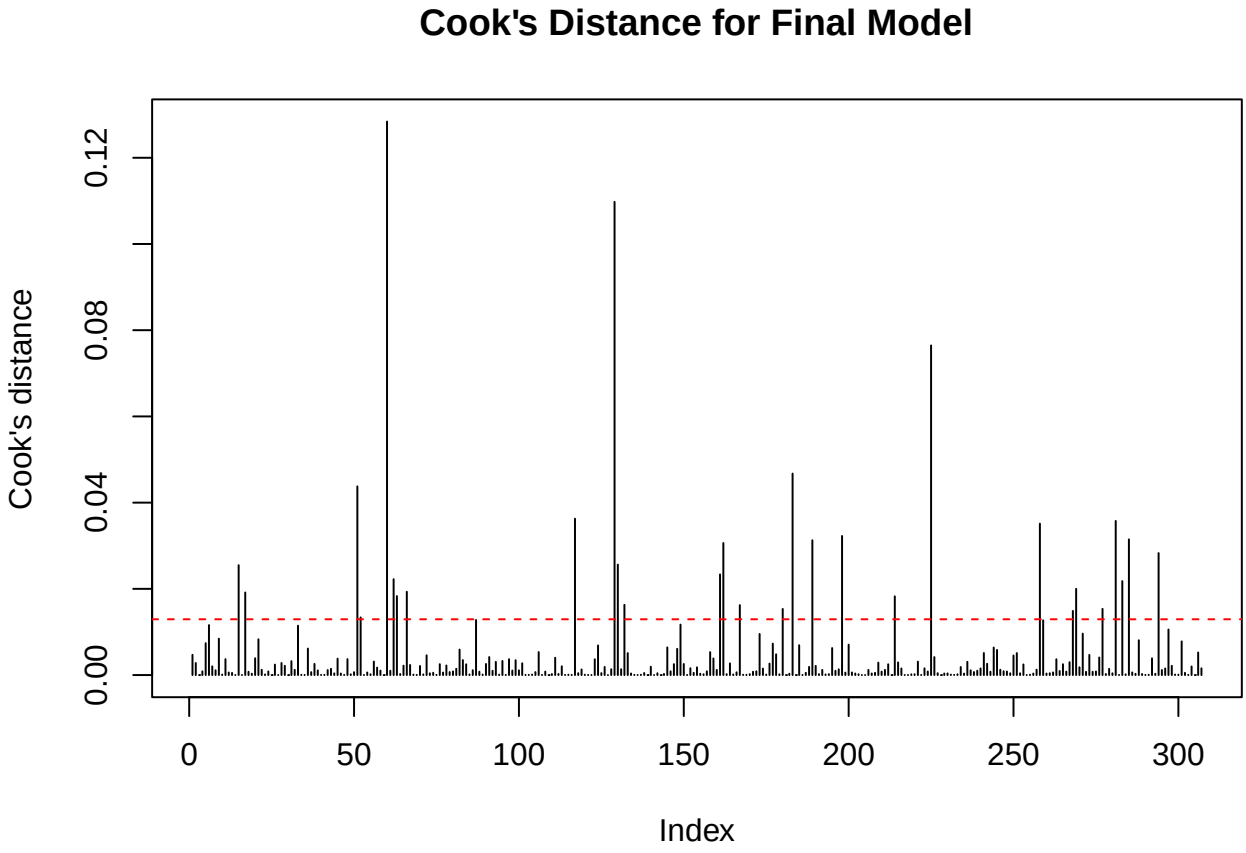


Figure A6

Cook's Distance for the Selected Regression Model



Code and data submission: Qianxue Wang